

Commodity Markets Intervention: Consequences of Speculation, and Informed Trading

Phat V. Luong ^{*1} and Ben Sopranzetti²

¹College of Business Management, SUNY Polytechnic Institute, Utica NY 13502

²Rutgers Business School, Rutgers University, Piscataway, NJ 08854

Abstract

The Chinese regulatory authority's interventions into the futures market for steel during 2016 provide a rare opportunity to study the unintended consequences of interventions. Although hot rolled coil (HRC) and rebar are products with nearly identical components, their futures contracts trade separately. The regulatory authority intervened into the futures market for rebar but refrained from doing so for HRC. We document impacts of interventions measured by increased volatility, trading volume, and spreads in rebar markets. There is also evidence of informed trading in both markets, reflected by abnormally high trading volumes and low open interests occurring the day before the announcement of interventions.

Keywords: Government intervention; speculation; informed trading; market microstructure; steel futures market.

Declarations of interest: None.

*Corresponding author: SUNY Polytechnic Institute, College of Business Management, 100 Seymour Rd, Donovan 1101, Utica NY 13502, e-mail: luongp@sunypoly.edu

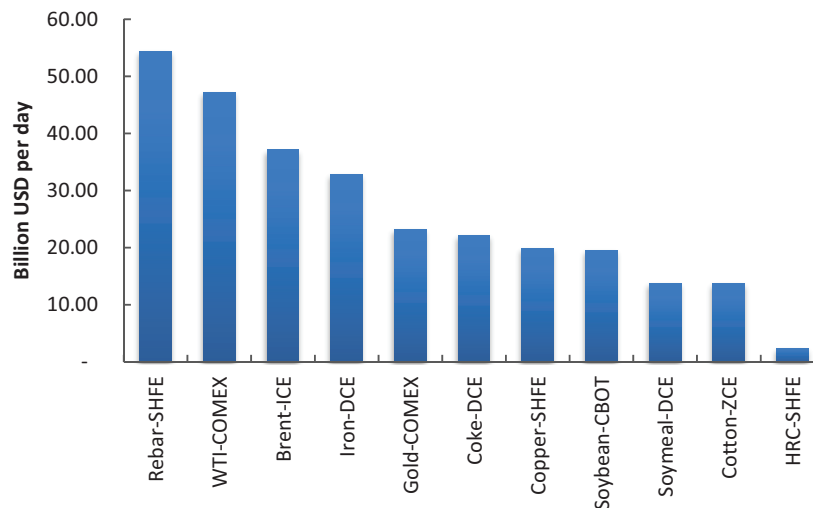
1 Introduction

Government and non-government regulatory agency intervention into the workings of financial markets can sometimes be massive, impactful events to quickly stabilizing markets, such as the short-selling ban in the U.S. during the 2008 financial crisis or China's sweeping clampdown of its stock markets during the 2015 market meltdown. Alternately, they can be smaller and implemented more regularly to subtly direct markets toward a government's goals, such as trading in the foreign exchange market to control exchange rates. Interventions are usually well-intended and driven with a particular goal in mind, but can sometimes result in unintended consequences that harm, rather than improve, market quality and the informational efficiency of asset prices in the affected market (Battalio et al., 2012; Pasquariello, 2018). Although there is active literature that examines the impact of government interventions on financial markets, the issue of causality has posed a challenge for researchers, due to the lack of a comparable control group when making pairwise comparisons between affected and unaffected market constituents. In the case of the 2008 U.S. short-sell ban, the effects of the short-sell ban were drawn from the comparison between stocks subject to the short-selling ban and those not subjected to the ban (Boehmer et al., 2013; Brogaard et al., 2017). However, the stocks in the banned group and the non-banned group represented different companies and have different underlying fundamentals. Consequently, there is potentially more to be learned from the use of a better-constructed comparison sample for the investigation of how governmental intervention can bleed across financial markets. To this end, the current paper examines the effects of the Chinese regulatory authority's intervention into its futures market; specifically, the market for steel.

There are currently two steel contracts listed on the Shanghai Futures Exchange (SHFE): reinforced bar (rebar) and the hot rolled coil (HRC) steel. Both contracts are offered with one-year expiration dates. The two contracts have different liquidity: the rebar contracts are traded ten times more often than HRC contracts, with a total volume of approximately 4 million compared to 300 thousand, respectively. Recently, rebar futures contracts surpassed West Texas Intermediate Crude Oil to become the most traded commodity futures contract in 2016 (see Figure 1), thus it is an important contract and worthy of investigation. Although the liquidity between rebar and HRC markets might be different, the underlying fundamentals are similar. The starting materials of both rebar and HRC are semi-finished casting products, such as steel slabs, blooms, and billets. Modern steel mills with dual capacities can easily switch from one type of product to the other. Moreover, their production growth rates are also very similar; for example, 29% and 26% in 2008, and 8% and 9% in 2009,

10% and 12% in 2014 for rebar and HRC respectively.¹ The returns of rebar and HRC spot prices on Shanghai Metal Market are highly correlated with Pearson's correlation of 0.500 with a significant p -value of near 0. As a result, the supply and demand conditions for rebar and HRC steel are more comparable than any other pairwise comparison made in previous studies.

Figure 1: Top 10 global commodity contracts by trading value, April 1 - April 29, 2016



Source: Bloomberg

In 2016, China experienced a massive commodities price bubble, with soaring prices and a tremendous increase in volatility, accompanied by (and perhaps even driven by) an increase in the number of speculative traders in commodity markets. Since rebar futures were the most heavily traded commodities, their market was subjected to substantial intervention during the 2016 commodity bubble. The interventions included a wide array of policy tools ranging from trading quotas, raising fees and trading margins, and cutting the number of new positions allowed each day. Traders and investors who violated these policies were temporarily blocked or suspended from trading. These interventions were not one-time in nature, unlike the short-sell ban in the U.S.; instead, they were dynamic, changing frequently (sometimes weekly). In contrast to the rebar futures market, the HRC futures market was spared from intervention, even though the two types of carbon steel are made from the same material. The use of contracts on rebar and HRC allows this paper to a great extent circumvent the challenges of making an apples to apples comparison encountered by prior

¹World Steel Association. Steel Statistical Yearbooks 2016. <https://www.worldsteel.org/en/dam/jcr:4f060d8b-3602-4ffe-9e87-7e93e0659449/Word+Steel+in+-Figures+2016.pdf>

papers when studying the impacts of government intervention.

To this end, this paper uses the official announcement of intervention activity on the SHFE to study the impact of the intervention policy on both the rebar and the HRC market. The results demonstrate that interventions in the rebar market worsened market quality for both the rebar and HRC market. Specifically, increased volatility was observed in both markets whenever news about interventions was published. However, the deterioration of other important market qualities, such as price impact, and bid-ask spreads was only present in the rebar market. In addition, the rebar market, but not the HRC market, experienced jumps the position tenure rate (PTR) whenever interventions were announced.

Interventions in the rebar market seem to have a spillover effect onto the HRC market, manifesting primarily as higher volatility. Moreover, an increased comovement between the two contracts one day before the announcement of intervention raises the suspicion of informed trading by insiders, who are taking advantage of their inside information by acting ahead of the intervention announcements. To provide evidence on the extent of informed trading in the HRC market related to intervention announcements in the rebar market, we test the ability of the trading activities to predict the probability of intervention. Interestingly, abnormally high trading volume and lower open interest at time t are statistically significant indicators of the intervention at time $t+1$.

The remainder of the paper are organized as follows. Section 2 provides an overview of the steel futures market of China and around the world. Section 3 discusses intervention policies and their impacts that have been documented in the literature. Section 4 presents the Madhavan et al. (1997) methodology, and estimates the various market quality factors such as price impact and implied bid-ask spread. Sections 5 analyses the impacts and the mechanism of the interventions. Finally, Section 6 concludes and provides policy implications.

2 Overview of the steel spot and futures markets

Steel is widely used in many industries including construction, automobiles, machinery, appliances, energy, and more. It is produced in many countries globally and is a heavily traded commodity. Global crude steel production has been growing for most of the last decade. Total steel production was 1.1 billion metric tons in 2005 and grew remarkably to 1.6 metric tons in 2015.² The underlying asset value of the global steel industry is enormous with an estimated value of about \$420bn compared to \$162bn of base metals and is growing sub-

²World Steel Association, World Steel in Figure 2016.<https://www.worldsteel.org/en/dam/jcr:4f060d8b-3602-4ffe-9e87-7e93e0659449/Word+Steel+in+Figures+2016.pdf>

stantially.³ Nevertheless, for many years, steel remained one of the few major commodities without a futures contract. With the rising demand, the price of the HRC - the most common steel product - rose by 195% from 2000 to its peak in 2008. Since steel is usually sold directly by steelmakers to large customers or through middlemen such as metal service centers, steel buyers have fewer mechanisms to manage their risks during periods of price volatility. That has left many buyers to rely on less-effective means to manage risk such as using a basket of stocks, bonds, derivatives and commodities that roughly tracks steel prices.

In response to the growing calls for a better risk management tool for steel, many major commodity exchanges planned to establish steel futures markets. The New York Mercantile Exchange (NYMEX) is the pioneer and the most vocal supporter of a steel futures market. In 2007, Robert Levin, senior vice president of the NYMEX, announced planned to launch a regional, cash-settled contract for US hot-rolled coil (HRC) based on SteelBenchmark by the end of the year.⁴ However, before NYMEX was able to launch its first steel product, NYMEX was absorbed by the Chicago Mercantile Exchange (CME). After acquiring NYMEX, CME started to offer its U.S. Midwest domestic HRC steel futures contract in October 2008. Shortly thereafter, the London Metal Exchange (LME), in April 2008, launched two physical-delivered futures contracts for steel billet: the Mediterranean with an initial delivery point in Turkey, and the Far Eastern with an initially delivery point in South Korea. Trading activity on Mediterranean contract soared 385% to 57,606 lots in the first six months of the year, equivalent to 3.74 million tonnes of steel, accounting for around 0.6% of the world's total billet production in 2009. Despite of the uptick at the beginning, the LME billet contract failed largely due to problems associated with storing and withdrawing steel from backlogged warehouses. These contracts were merged to create a global contract under the Mediterranean contract in July 2010. LME later in November 2015 offered two new cash-settled contracts: one for steel reinforcing bar and one for steel scrap. Quickly joining the pack, the Shanghai Futures Exchange (SHFE) began offering two construction steel products: reinforcing bar and wire rod in April 2009. In 2014, SHFE introduced a hot-rolled coil contract.

Among the steel futures markets, SHFE has traditionally been the most active. However, an issue arose when the influx of day traders headed to commodity markets to search for higher returns. The total volume of steel futures on SHFE surpassed all shares traded on China's equity markets in April 2016.⁵ The explosion of trading activity and the ensuing

³Trade statistics for international business development, Trade Map 2015. <https://www.trademap.org/Index.aspx>

⁴American Metal Market, American Metal Market's Guide to Steel Futures. http://www.steelbenchmarker.com/files/steel_futures.pdf

⁵Henry Sanderson, Speculators march into China commodities, Financial Times. <https://www.ft.com>

jump in volatility prompted China's authorities to scramble to prevent markets from melting down. The goal of the interventions was to bring down speculative trading activity and volatility.⁶ It is common for the Chinese government to actively manage its financial markets to promote financial stability. They stepped in to calm down the turmoil in their stock markets in 2015 by buying back stock, lowering interest rates, imposing a moratorium on initial public offerings, and even suddenly devaluating the yuan. These interventions raised doubts about the government's ability to manage its overheating financial system and have been a great source of anxiety for investors and policymakers around the globe. When intervention started into the rebar futures market, it began with the investigation and punishment of the traders with abnormal trading activities on March 7, 2016. Nevertheless, the market conditions continued to worsen leading to the decision to raise trading fees across markets on April 21, 2016. This action could potentially have effects on both markets, so to address this later in this paper, we examined the impacts of interventions on both markets with and without this date but found no difference in our results. Simultaneous to the interventions that were occurring in the futures markets, there was a government crackdown on high-frequency trading from March to late July 2016, where traders were blacklisted if they opened and closed positions too frequently daily. Further, they could be blocked from trading for one month. The intention was to slow down the pace of trading activities. However, despite these well-intentioned interventions, trading volume continued to spike. On June 1, 2016, SHFE announced that it would enforce a more draconian quota on the number of transactions and would close any position that exceeded the quota limit. The crackdowns on traders seemed to happen only in the rebar market. That could be due to the fact that the rebar market was much more liquid than the HRC market (which might permit speculative traders to disguise their trades and easily exploit the market. Nevertheless, we document all of the intervention announcements on SHFE and investigate their mechanism and impacts on the trading activities and market quality in the later sections.

3 Literature review

The impacts of government and non-government regulatory intervention on asset prices and the macroeconomy are well documented in the literature. However, studies on the effects of interventions in financial markets are quite limited. The extent of these studies is confined to either static policies such as the short-sale bans or dynamic intervention by participating

/content/592c8af4-0b94-11e6-b0f1-61f222853ff3

⁶Xiaowen Bi, Exclusive: China securities regulator confirms order to curb futures speculation, Reuters. <https://www.reuters.com/article/us-china-commodities-regulator/exclusive-china-securities-regulator-confirms-order-to-curb-futures-speculation-idUSKCN0XP2EQ>

in the market such as government trading in the foreign exchange market.

The intervention into financial markets that has received the most attention recently is the short-sale ban implemented by the US Securities and Exchange (SEC) in 2008. On July 15, 2008, the US SEC placed temporary restrictions on naked short sales of the stocks of 19 financial firms. Boulton and Braga-Alves (2010) showed that naked short sales for the restricted stocks were successfully eliminated but naked short sales increase dramatically for a closely matched sample of financial firms during the restriction period. They also found that restrictions negatively impacted various measures of liquidity, including bid-ask spreads and trading volume. In September 2008, the US SEC decided to temporarily disallow short-sales of nearly 1,000 financial stocks. Using proprietary NYSE data, Boehmer et al. (2013) constructed a difference in difference tests by comparing 200 stocks on the ban list to 1,066 non-banned stocks to examine the impacts of the ban. Boehmer et al. (2013) showed that the market quality of stocks subjected to the ban deteriorated severely. The short-sale ban was shown to be associated with larger bid-ask spread and higher intraday price volatility. Brogaard et al. (2017) raised the concern that the ban could be correlated directly with changes in liquidity unrelated to short selling since the stocks selected for the ban in Boehmer et al. (2013) were not randomly selected. Thus, to mitigate the concern, Brogaard et al. (2017) examined the effects of the ban on high-frequency trading (HFT) and non-HFT using NASDAQ data. They found that HFTs' trading and HFT's short-selling harm liquidity by adversely selecting liquidity suppliers while non-HFTs' short selling activity improves liquidity. The ban has largely eliminated HFTs' shorting activity but it has a smaller impact on the overall HFT activity. However, Brogaard et al. (2017) doubted that limiting HFTs' ability to demand liquidity might not improve overall market liquidity since it could impair HFTs' ability to manage risk and thereby supply liquidity. The short-selling ban not only had detrimental effects on the banned stocks but also had adverse effects on the options market. The ban caused option trading costs to increase sharply and lead to a bias in the relative prices of options and stocks on the ban list. Félix et al. (2016) studied the short-sale ban on the European market in 2011. It found that the ban negatively impacted the affected stocks by increasing implied jump risk levels. It also observed the shift in short positions from the banned stocks to the non-banned ones.

Another policy that was often used to stabilize the market is the setting of a daily price limit. However, this tool has been shown to have a counterproductive effect. Chen et al. (2019) showed that the setting of a daily price limit induced large investors on the Shenzhen Stock Exchange to pursue a destructive strategy of pushing stock prices up to the upper limit and then selling on the next day.

Besides setting policies, governments can also participate directly in the market to steer

the market in the desired direction. Central banks around the world occasionally intervene in the foreign exchange markets to control the exchange rates. Naranjo and Nimalendran (2000) showed that government interventions created significant adverse selection problems for dealers. They found that the bid-ask spread of the DM/US\$ exchange rate increased with the intervention during the period 1976-1994. This effect only occurred when the interventions were unexpected. Government interventions affect not only the trading cost but also the attention to market information. Peiers (1997) used Reuters screen headlines of central bank intervention as an information trigger to identify asymmetric information flows and price leadership patterns in the DM/US\$ exchange market. The results revealed that Deutsche Bank was treated as a price leader up to 60 minutes prior to the intervention report. By the time of the Reuters announcement, traders resumed their preintervention trading interactions. Harvey and Huang (2002), using data on open market operations from 1982-1988, showed that the Federal Reserve Bank's trading on both fixed income instruments and foreign currencies dramatically increased the volatility. Pasquariello (2007) found that Swiss National Bank interventions failed to smooth CHFUSD return fluctuations. Moreover, Swiss National Bank interventions induced significant misinformation and heterogeneity of beliefs among market participants and deteriorated market liquidity. Pasquariello (2018) also reported evidence of violations of the law of one price in arbitrage-related markets induced by direct government intervention in the market.

Brunnermeier et al. (2017) presented a unified theoretical framework for how China might manage its financial system, whereby government intervention aimed at preventing a market breakdown and an explosion of volatility could alter market dynamics by introducing noise that drives asset prices and that divert investor attention toward acquiring information about this noise factor rather than fundamentals. Hence, government intervention could worsen the informational efficiency of asset prices. Draper and Fung (2003) show that the direct market intervention by the Hong Kong government in both the stock and futures markets during the period of the Asian financial crisis not only affected the pricing relationship between the Hang Seng Index futures and the cash index at that moment but also had a lasting impact that continued after the intervention ended. Their results show that arbitrage efficiency was impeded during and in the immediate aftermath of, the intervention.

4 A structural model of price formation

In order to examine the impact of government intervention on market quality, we first need to estimate the key market quality factors such as the price impact and the bid-ask spread. We do so by examining SHFE tick-by-tick data using the well-established microstructure

model of Madhavan, Richardson, and Roomans (1997) (MRR). The MRR model provides insights into the determinants of the autocorrelations of quotes and returns as well as the price impact and the variance of bid ask spread and order flow.

4.1 MRR model

Consider the market for a risky security whose fundamental value evolves through time. The security is traded in an auction-dealer mechanism where liquidity providers quote bid and ask prices at which they are willing to trade. An order also may be executed within the quotes. Let p_t denote the transaction price of the security at time t . x_t denotes an indicator variable for trade initiation, where $x_t = +1$ if trade is buyer initiated and -1 if the trade is seller initiated. Denote by ε_t the innovation in beliefs between time $t-1$ and t due to new public information. ε_t is assumed to be an independent and identically distributed random variable with mean zero and variance σ_ε^2 . MRR assumes that traders have private information that can affect fundamental asset values, which would otherwise follow a random walk. Let μ_t denote the post-trade expected value of the stock conditional upon public information and the trade initiation variable. The revision in beliefs is the sum of the change in beliefs due to new public information and order flow innovation.

$$\mu_t = \mu_{t-1} + \theta (x_t - E[x_t|x_{t-1}]) + \varepsilon_t \quad (1)$$

If market makers believe that some traders possess private information about fundamental asset value, a buy (sell) order is associated with an upward (downward) revision of beliefs. The change in beliefs due to order flow is $\theta (x_t - E[x_t|x_{t-1}])$, where $(x_t - E[x_t|x_{t-1}])$ is the surprise in order flow and $\theta \geq 0$ measures the degree of information asymmetry or the so-called permanent impact of the order flow innovation. Higher values of θ indicate larger revisions for a given innovation in order flow.

Market maker bid and ask quotations are ex post rational so that the ask (bid) price is conditioned on a trade being buyer initiated (seller initiated). Let p_t^a , p_t^b denote the (pre-trade) ask price, and bid price at time t respectively. Let $\phi \geq 0$ represent the market maker's cost per share for supplying liquidity. We can interpret ϕ as the dealer's compensation for transaction costs, inventory costs, risk bearing, and possibly the return to their unique position.

$$p_t^a = \mu_{t-1} + \theta (1 - E[x_t|x_{t-1}]) + \phi + \varepsilon_t \quad (2)$$

$$p_t^b = \mu_{t-1} - \theta (1 + E[x_t|x_{t-1}]) - \phi + \varepsilon_t \quad (3)$$

θ captures the temporary (transitory) effect of order flow on prices. Orders can be executed at the quoted bid or ask price, and sometimes within the bid-ask spread. In the later case, transactions are assumed to be executed at the midquote, $(p_t^a + p_t^b)/2$. Then the transaction price can be expressed as

$$p_t = \mu_t + \theta x_t + u_t \quad (4)$$

where u_t is an independent and identically distributed random variable with mean zero. The term u_t captures the effect of stochastic rounding errors included by price discreteness or possibly-time varying returns. Any systematic deviation from zero in the rounding error is captured in the dealer cost component ϕ . Dealer must incorporate the transaction cost ϕ into the quotes. Therefore, the transaction price at time t has already impounded the effect of the trade direction on the incoming trade.

$$p_t = \mu_{t-1} + \theta (x_t - E[x_t|x_{t-1}]) + \phi x_t + u_t + \varepsilon_t \quad (5)$$

MMR assumes that the trade initiation variable follows a Markov process. The probability that a transaction at the ask(bid) follows a transaction at the ask (bid) is

$$\gamma = \Pr[x_t = x_{t-1} | x_{t-1} \neq 0] \quad (6)$$

The first order autocorrelation of the trade initiation variable is $\rho = E[x_t|x_{t-1}]/Var[x_{t-1}] = 2\gamma - (1 - \lambda)$, with λ is the probability of a midquote execution. When there is no possibility of transacting within the quotes, $\lambda = 0$. Then the conditional expectation of the trade initiation variable given public information are

$$E[x_t|x_{t-1} = +1] = \gamma - (1 - \gamma) = \rho \quad (7)$$

$$E[x_t|x_{t-1} = -1] = (1 - \gamma) - \gamma = -\rho \quad (8)$$

thus the conditional expectation $E[x_t|x_{t-1}] = \rho x_{t-1}$.

Equation 5 can be expressed as

$$p_t - p_{t-1} = (\phi + \theta) x_t - (\phi + \rho\theta) x_{t-1} + \varepsilon_t + u_t - u_{t-1} \quad (9)$$

$$\Delta p_t = (\phi + \theta) x_t - (\phi + \rho\theta) x_{t-1} + \xi_t \quad (10)$$

where $\xi_t = \varepsilon_t + u_t - u_{t-1}$ is a composite of public information and microstructure noises.

ξ_t is assumed to be an independent and identically distributed random variable with mean zero and variance σ_ξ^2 .

The parameters governing the behavior of transaction prices and quotes, the asymmetric information parameter, θ , the cost of supplying liquidity, ϕ , and the autocorrelation of the order flow ρ , can be estimated by the generalized method of moments (GMM) with the population moment conditions implied by the model

$$E \begin{pmatrix} x_t x_{t-1} - x_t^2 \rho \\ \xi_t \\ \xi_t x_t \\ \xi_t x_{t-1} \end{pmatrix} = 0 \quad (11)$$

where $\xi_t = \Delta p_t - (\phi + \theta) x_t - (\phi + \rho\theta) x_{t-1}$. The first equation is the definition of the autocorrelation in trade initiation. The second equation defines the pricing error. the last two equation are the OLS normal equations.

The implied bid-ask spread is $2(\theta + \phi)$ and the price impact for a buyer initiated trade is $\theta(1 + \rho)$.

4.2 Parameter estimates

Data

We use the most traded contracts, those with an October 2016 expiration (RB1016 and HC1016), to examine the evolution of the information parameters over the trading lives of these contracts. Both the rebar and HRC contracts have a one-year expiration. The tick data for RB1016 is from 10/30/2105 to 9/30/2016 and has 1629937 observations. The HC1016 starts from 1/3/2015 to 9/30/2016 and has 1034237 observations. The data for rebar and HRC steel contracts could be obtained from Bloomberg. The trades were classified as buy or sell based on the tick rule, which categorized trades based on the price movements. When the trade price is higher (lower) than the previously recorded price, the trade is classified as a buy (sell) initiated trade. If there is no price movement, that is, the current price and the previous price are the same, then the trade is classified to the same category as of the last trade. The volatility was estimated based on one-minute price data.

Since the SHFE trading hours are from 9:00 to 11:30 and from 21:00 to 3:00 of the next day, the data would be divided into two periods to avoid distortion in estimates due to new information arriving during the break that could cause prices to jump. We call the trading period from 9:00 to 11:30 the “morning session,” and the period from 21:00 to 3:00 the “night session.” To ensure there are sufficient observations for model estimation, only

trading sessions that have at least 100 trades per period are considered.

Parameter estimates

Table 1 and 2 present summary statistics on the individual parameter estimates governing the stochastic process for the transaction price changes of rebar and HRC contracts with October 2016 expiry on the SHFE. The tables present the mean coefficient estimates and the mean standard errors of the estimated parameters for these two contracts in a one-month interval for the eleven months leading to the expiry of them. The daily values will be the average of the morning and the night sessions since there is no significant difference between the estimates of these two trading sessions for each contract. It is noticeable that all of the standard errors are relatively small compared to their estimates showing the reliability of these estimates. J -test statistics also assures that the model is not overidentified.

The information asymmetry, θ , is the main parameter of interest. From table 1, it is clear that the degree of asymmetry for the rebar contract decreased for the first four months of its life before increased from March through July, then decreased again for the rest of the life of the contract. On the other hand, the θ of the HRC contract continually dropped over three quarters of its life as trading activities increased and only climbed back up when trading was fading (see Table 2). θ represents the magnitude of the revision in the market maker's beliefs concerning the security's value induced by order flow. A decline in θ represents less reliance on the signal content of order flow. The lesser reliance on prior beliefs is consistent with either market makers not learning about fundamental asset value through the trading process, or the presence of a smaller percentage of liquidity traders.

The transaction cost ϕ increased over the life of the rebar contract. It increased an average of 19 cents from October to July, a rise of about 46 percent, before it slightly decreased. On the contrary, the transaction cost of the HRC contract decreased gradually throughout its life: on average, the trading cost of the morning session and the night session decreased 94 cents from 1.52 to 0.58 and 15 cents from 0.70 to 0.55 respectively from January through September. The transaction cost ϕ represents the economic cost of market making, and the increase in this parameter reflects the increasing risks associated with carrying inventory.

The implied spreads, S^{MRR} , should exhibit the familiar U-shape pattern over the life of the contracts. However, the spread of rebar spiked up from March through July. r is the fraction of the implied spread attributed to asymmetric information. If $r = 0$, the spread is entirely attributable to the costs of supply liquidity, and if $r = 1$, direct liquidity costs are negligible and adverse selection costs constitute the entire bid ask-spread. The rising r of rebar from March through July indicates the higher costs of adverse selection.

Next, we control for the difference in volume and volatility throughout the contract life

Table 1: The GMM model parameter and trading cost estimates of rebar futures contract expired 10/2016

	θ	ϕ	ρ	J -test	S^{MRR}	r	Price Impact
November	0.137 (0.002)	0.437 (0.026)	0.690 (0.024)	1.377 (0.450)	1.1486 (0.0021)	0.2367 (0.0009)	0.2332
December	0.098 (0.011)	0.442 (0.011)	0.588 (0.022)	1.564 (0.455)	1.0816 (0.0003)	0.1811 (0.0003)	0.1553
January	0.097 (0.009)	0.453 (0.009)	0.501 (0.023)	1.613 (0.398)	1.1011 (0.0002)	0.1754 (0.0001)	0.1452
February	0.082 (0.007)	0.469 (0.006)	0.348 (0.022)	2.063 (0.407)	1.1013 (0.0001)	0.1478 (7.83e-05)	0.1099
March	0.221 (0.012)	0.549 (0.009)	-0.051 (0.016)	4.439 (0.322)	1.5411 (0.0010)	0.2710 (3.28e-05)	0.2062
April	0.280 (0.014)	0.594 (0.010)	-0.109 (0.015)	8.718 (0.186)	1.7490 (0.0013)	0.3002 (3.92e-05)	0.2505
May	0.233 (0.013)	0.570 (0.010)	-0.109 (0.015)	8.848 (0.190)	1.6064 (0.0010)	0.2864 (2.81e-05)	0.2068
Jun	0.183 (0.011)	0.544 (0.008)	-0.134 (0.015)	4.071 (0.272)	1.4531 (0.0007)	0.2475 (2.86e-05)	0.1586
July	0.305 (0.016)	0.610 (0.017)	-0.116 (0.015)	7.846 (0.109)	1.8302 (0.0013)	0.3302 (3.43e-05)	0.2694
August	0.197 (0.011)	0.547 (0.009)	-0.069 (0.016)	3.868 (0.276)	1.4882 (0.0007)	0.2598 (3.43e-05)	0.1807
September	0.173 (0.014)	0.492 (0.017)	0.377 (0.023)	1.876 (0.438)	1.3314 (0.0015)	0.2583 (0.0002)	0.2395

This table reports the monthly average of daily GMM parameter estimates of the MRR model for the rebar contract with October 2016 expiry. GMM standard errors are in the parentheses.

and regress the intervention announcements on information asymmetry and transaction costs according to the following specification,

$$Y_t = \alpha + \beta Int_t + \ln(Volume_t) + \ln(Vol_t) + \varepsilon_t \quad (12)$$

where $Y_t = \{\phi, \theta\}$, and Int_t is the intervention indicator which takes value 1 if there was an intervention announcement at day t and 0 otherwise. The results in the Table 3 confirms our observations.

Moreover, the autocorrelation of order flow ρ of rebar contract gradually declined and became negative from March to August. The negative sign of order flow autocorrelation suggests that traders liquidated their positions when price moved up (down) rather than piling up inventory or holding on to their positions. The negative autocorrelation of order

Table 2: The GMM model parameter and trading cost estimates of the HRC futures contract expired 10/2016

	θ	ϕ	ρ	J -test	S^{MRR}	r	Price Impact
January	1.023 (0.447)	0.838 (0.592)	0.794 (0.054)	0.598 (0.560)	3.889 (0.733)	0.640 (0.155)	1.851
February	0.624 (0.188)	0.770 (0.295)	0.775 (0.035)	0.394 (0.655)	2.789 (0.383)	0.463 (0.033)	1.114
March	0.461 (0.054)	0.585 (0.073)	0.608 (0.024)	1.168 (0.444)	0.463 (0.021)	0.441 (0.001)	0.744
April	0.343 (0.019)	0.582 (0.005)	0.188 (0.019)	3.932 (0.295)	1.851 (0.003)	0.361 (0.0001)	0.384
May	0.286 (0.015)	0.566 (0.004)	0.090 (0.019)	3.004 (0.343)	1.705 (0.002)	0.334 (4.34e-05)	0.3111
Jun	0.232 (0.014)	0.542 (0.002)	0.136 (0.020)	1.626 (0.438)	1.549 (0.001)	0.298 (5.40e-05)	0.262
July	0.352 (0.018)	0.632 (0.003)	0.031 (0.018)	3.657 (0.229)	1.969 (0.002)	0.356 (4.62e-05)	0.361
August	0.256 (0.015)	0.554 (0.002)	0.164 (0.021)	1.665 (0.380)	1.620 (0.001)	0.315 (4.92e-05)	0.296
September	0.333 (0.052)	0.549 (0.064)	0.561 (0.026)	0.754 (0.527)	1.764 (0.038)	0.369 (0.002)	0.538

This table report the monthly average of daily GMM parameter estimates of the MRR model for the HRC contract with October 2016 expiry. GMM standard errors are in the parentheses.

Table 3: Impacts of interventions on the information asymmetry and market making costs

	Rebar	HRC
Market making cost ϕ	0.017** (0.005)	-0.035 (0.043)
Information asymmetry θ	0.045*** (0.009)	0.021 (0.036)

This table reports the coefficients of the interventions on the information asymmetry and transaction cost of the rebar and HRC contracts. The estimates are on the top rows, the standard errors are in the parentheses on the bottom rows. The “***” represents the significant level at the 0.1%, “**” at the 1%, “*” at the 5%, and “.” at the 10%.

flow reflects higher transaction costs and higher risk of carrying inventory as suggested by Table 3. For the HRC contract, ρ is positive and has a U-shape pattern. This suggests that order flow plays an important role in determining the direction of the price of the HRC contract.

The price impact of trades for the rebar contract initially decreased from December through February, and then increased when the interventions were rolled out. In contrast, the price impact of the HRC contract had a typical U-shaped pattern.

The implied bid-ask spreads of the HRC contracts that we examined exhibit a normal U-shaped pattern over their lives. This reflects the illiquidity at the beginning and the end of the HRC contract. In contrast, the implied bid-ask spread of the rebar contract has an inverse U-shape which implies higher adverse selection costs. The higher adverse selection cost implies the presence of relatively more informed traders. Hence, market makers set relatively higher spreads to compensate themselves for losses caused by adverse selection. It is noticeable that the market making cost, ϕ , is a dominating factor in determining the bid-ask spread of the rebar contract. Contrarily, transaction costs and information costs equally contribute to the determination of the bid-ask spread of the HRC contract.

5 Mechanisms and Impacts of interventions

We begin by examining how trading activities and market quality react to intervention announcements. Thereafter, the interaction between intervention announcements and trading activities and market quality factors will then be further investigated to better understand the mechanisms through which intervention makes its impacts.

5.1 Effects of intervention announcements on trading activities and market quality

We begin by examining whether trading activity or market quality changes on the days where there are intervention-related announcements. These interventions include all of the interferences described in Section 2. To this end, we regress the trading activity and market quality factors on an intervention dummy variable, Int_t , which equals one for the days when an intervention was announced and zero otherwise.

$$Y_t = \alpha + \beta Int_t + \ln(Volume_t) + \ln(Vol_t) + \varepsilon_t \quad (13)$$

where Y_t is the trading activity or market quality factors at time t , and $Int_t = 1$ for the days of the intervention announcements and zero otherwise. Please note that volume and volatility are of course not control factors for the estimate of the intervention effect on themselves, respectively. The intervention announcements are documented based on the official news and announcements on the SHFE website. The estimates of intervention impacts are presented in Table 4

Table 4: Effects of intervention announcements

	Rebar	HRC
Volume	0.239 (0.272)	0.234 (0.407)
PTR	0.280 (0.158)	0.012 (0.724)
PTA	0.055* (0.027)	0.050 (0.046)
Volatility	4.860*** (0.853)	4.541*** (1.123)
Price Impact	0.047*** (0.010)	0.015 (0.065)
Implied Bid-Ask Spread	0.124*** (0.025)	-0.028 (0.094)

This table presents the estimates of the intervention effects from model 13. The estimates are on the top rows, the standard errors are in the parentheses on the bottom rows. The “***” represents the significant level at the 0.1%, “**” at the 1%, “*” at the 5%, and “.” at the 10%.

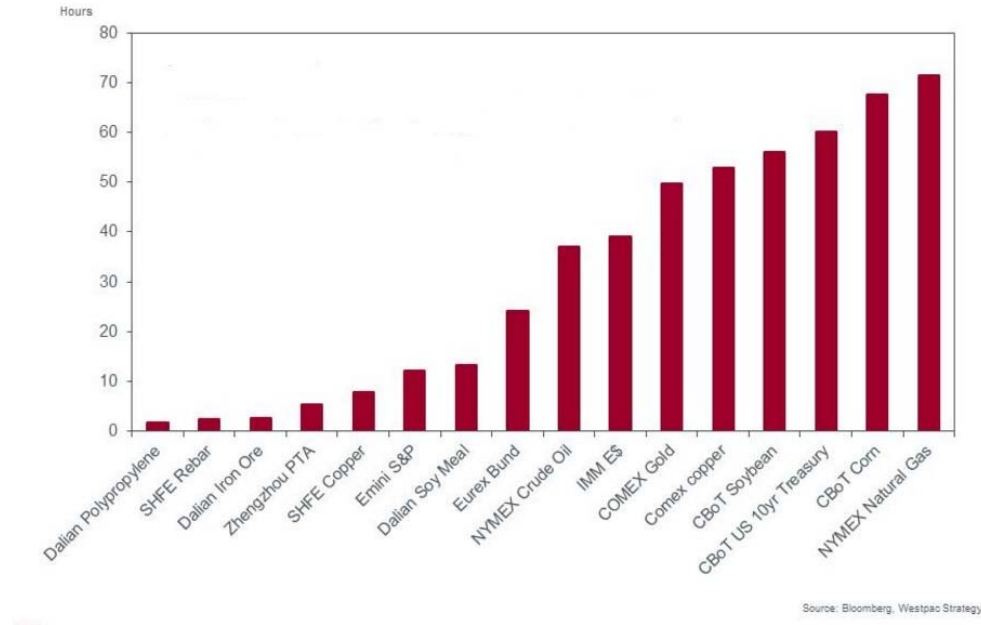
Impacts on trading activities

Table 4 shows that the interventions do not have a significant impact on the trading volume of either contract and seem not to be able to reduce the trading activity as intended. This result is not consistent with that reported in Boehmer et al. (2013) regarding abnormal trading volume in the banned stock in the New York Stock Exchange during the 2008 short-sale ban. The increase in the trading volume of the banned stocks on the NYSE was considered to be the effect of bailout news.

Increasing information asymmetry and inventory holding costs are potential reasons why traders are less likely to hold on to their positions, as illustrated by the decrease in the position tenure rate associated with the intervention announcements. The tenure of the contract position is a commonly used proxy in practice for the level of speculation in a market.⁷ The tenure is calculated as trading hours*(average aggregate open interest/average aggregate volume). A lower tenure indicates that, on average, positions are held for a shorter period of time, and a lower average tenure is associated with an increase in speculation. The average tenure of a rebar contract is less than 3 hours compared with 40 hours for WTI crude, 60 hours for copper, and 70 hours for natural gas.

⁷David Scutt, Chinese commodity futures are tanking, including iron ore, Business Insider Australia. <https://www.businessinsider.com.au/chinese-commodity-futures-are-tanking-including-iron-ore-2016-5>

Figure 2: Average position tenure for major futures contracts



Source: Bloomberg, Westpac Strategy

Since rebar and HRC share the same trading hours, we can directly compare their position tenure rates, as will be documented below. Position tenure rate (PTR) is the ratio of the aggregate open interest over the aggregate volume and is associated with the likelihood of a contract being turned over. The lower the position tenure rate, the more likely the contract will be turned over in a given period of time, which in turn gives an indication of the level speculation in a market. When we do so, we only see the impact of interventions on the PTR of the rebar but the HRC.

Although position tenure rate is important, its behavior depends not only on actions of traders on the day t but also on the contract's whole history up to the same day. Lucia and Pardo (2010) suggest that the degree of speculation may be more precisely assessed by looking at the change of the aggregate open interest over the aggregate volume, or "position tenure acceleration" (PTA) as we call it. Thus, it is of greater interest to examine the PTAs of the two contracts. Similar to the PTR, interventions impact only the PTA of the rebar contract. As government intervening, the changes of positions in rebar contracts accelerate indicate speculation activities. That behavior is not observed in the HRC market.

Impacts on market quality

It has been documented that interventions increase volatility in a market (Boehmer et al., 2013), elevate the price impact of trades (Boehmer et al., 2013; Brogaard et al., 2017), increase bid-ask spreads (Naranjo and Nimalendran, 2000; Boehmer et al., 2013; Brogaard et al., 2017), and worsen the information efficiency of asset prices (Pasquariello, 2018). Table 4 provides evidence that confirms the existence of those impacts of government interventions on the futures market.

It is in the market quality factor where evidence of contagion across the two markets starts to appear. The volatility of both the rebar and the HRC rose when the interventions were revealed (see Table 4). By setting different position limit and trading quotas for different types of traders, or selectively suspending trader accounts, the regulator either intentionally or unintentionally gets into the business of deciding winners and losers. Due to the fear of unexpected interventions, traders may potentially avoid holding inventory or may seek alternative assets in which to invest. As a consequence, we see evidence of negative autocorrelations for order flow in the rebar contract and a spill-over of increased volatility in the HRC contract.

Price impact is the consequence of the market activity that is often overlooked in previous studies. Boehmer et al. (2013) was the only study that examined the consequence of the price impact of the short-sell ban, by investigating the effect of the ban on the price impact of both the banned and the non-banned group. The price impact of the banned group was found to increase by more than two times that of the non-ban group. Price impact measures the liquidity of the market due to the imbalance of supply and demand. This imbalance is caused by large buy (sell) orders or informed trading. In the case of steel, the increase in price impact was most likely caused by the informed trading, since the information asymmetry θ of rebar increases and autocorrelation of rebar order flow is negative.

The implied bid-ask spreads of both the rebar and HRC contracts show similar patterns. The spreads for rebar contract widened when the interventions were announced. This impact was not observed in the HRC contract.

The interventions not only worsened market quality but also diminished the informativeness of asset prices. Brunnermeier et al. (2017) argues that intervention introduces noise into asset prices, which in turn attracts speculative investors by diverting their attention away from asset fundamentals. Consequently, market participants have less incentive to acquire fundamental information, thus leading to less informative asset prices. We used a Chow test to examine whether the links between the futures contracts and their fundamentals, i.e., the spot prices on the Shanghai Metal Market, has been altered due to the interventions. The futures and spot prices we use are the daily price. For the period before interventions, the

rebar prices are from 11/17/2019 to 3/4/2016 and contain 1054 observations while the HRC prices go from 1/15/2015 to 3/4/2016 and has 271 observations. For the intervention period, both rebar and HRC start from 3/5/2016 to 4/28/2017 and each has 277 observations.

$$F_t = \alpha + \beta TR_t + \phi S_t + \varepsilon_t \quad (14)$$

where F_t is daily settlement prices of rebar and HRC futures contracts, S_t is the spot price, and TR_t is a trend variable. The unrestricted regressions allow the coefficients to be different in the two regimes and are delineated as follows:

$$F_{1,t} = \alpha_1 + \beta_1 TR_{1,t} + \phi_1 S_{1,t} + \varepsilon_{1,t} \quad (15)$$

and

$$F_{2,t} = \alpha_2 + \beta_2 TR_{2,t} + \phi_2 S_{2,t} + \varepsilon_{2,t} \quad (16)$$

We use the date, 3/7/2016, when the interventions started in the market to mark the boundary of the two regimes. The first regime is from the inception of each contract (excluded the first year to avoid lack of trading activities) to 3/7/2016, and the second regime is from 3/8/2016 to 6/1/2017. The null hypothesis is that $\alpha_1 = \alpha_2$, $\beta_1 = \beta_2$, and $\phi_1 = \phi_2$. The F test statistic can be computed as follow

$$\frac{(SSR - (SSR_1 + SSR_2)) / k}{(SSR_1 + SSR_2) / (N_1 + N_2 - 2k)} \quad (17)$$

where SSR , SSR_1 and SSR_2 are the sum of squared residuals from the restricted regression and the two unrestricted regressions, respectively, N_1 and N_2 are the number of observations in each regime, and k is the number of coefficients.

The test statistics for the shift in the relationship of the rebar and the HRC contracts with their fundamentals are 97.648 (p -value = 0.000) and 134.590 (p -value = 0.000), respectively. The p -values confirms the significance of the test statistics and verifies the shifts in the links of these contracts and their fundamental.

5.2 Informed trading

Big Chinese financial institutions and state-backed enterprises rely on their “public relations experts” to scour for information. “If you ask any Chinese investor whether they think a chairman or executive of a company is trading in their own stock, nine out of 10 would expect that to be the case just as they would expect a government official to be taking

bribes,” says Fraser Howie, co-author of *Red Capitalism*. Everyone appears to trade based on tips from contacts who claim to have insider information. Senior regulatory officials admit that government officials who have access to insider information through their duties and their relatives are some of the worst offenders.⁸

We will further investigate the extent to which traders know in advance information regarding the government’s interventions by examining the pattern of the trading volume and the open interest prior to intervention announcement days. Leaks give traders an opportunity to anticipate the market reaction and to take profitable trading positions. Hence, if there were informed trading due to government leaks, we would expect to see abnormal trading activities immediately prior to official announcements. Moreover, the existence of these abnormal trading activities would be predictive of government interventions. To test this possibility, we employ the following logit model

$$\begin{aligned} \text{logit}(Int_t) = & \alpha + \beta_1 \ln(Volume_{t-1}) + \beta_2 \ln(OI_{t-1}) + \beta_3 \ln(Vol_{t-1}) \\ & + \beta_4 \ln(Volume_{t-2}) + \beta_5 \ln(OI_{t-2}) + \beta_6 \ln(Vol_{t-2}) \end{aligned} \quad (18)$$

Table 5 presents the results of the analysis of whether abnormal trading activity a day before the intervention announcement could predict the upcoming intervention. Higher trading volume and lower open interest are associated with a higher likelihood of intervention. Investors with inside information seem to exit the market the day before the government takes action. Thus, the flow of money out of the market will be indicated by a high trading volume accompanied by low open interest.

Table 5: Predictability of past trading activities on future interventions

$\text{logit}(Int_t)$	α	$Volume_{t-1}$	OI_{t-1}	Vol_{t-1}	$Volume_{t-2}$	OI_{t-2}	Vol_{t-2}
Rebar	8.742 (14.292)	2.997* (1.239)	-7.051. (4.129)	0.483 (0.665)	-0.768 (1.278)	4.065 (3.746)	-1.165. (0.672)
HRC	2.968 (3.459)	0.962 (0.731)	-1.663 (2.790)	0.916 (0.621)	-0.761 (0.746)	0.991 2.630	-0.311 0.593

This table reports estimates of Eq.18. The dependent variable is the logit function of the intervention. Standard errors are in parentheses. “***” indicates significance at 0.1%, “**” at 1%, “*” at 5%, and “.” at 10% level.

⁸Jamil Anderlini, China starts crackdown on insider trading, Financial Times. <https://www.ft.com/content/751a82ca-f7f4-11df-8d91-00144feab49a>

6 Conclusion

The intervention into the Shanghai futures market for rebar steel, but not hot rolled coil steel (which shares many fundamental properties with rebar) provides a natural testing ground to examine the indirect, unintended, and contagious effects of interventions designed to impact trading in one market on another unintervened market. Evidence indicates that government intervention into the rebar market increased the position tenure rate, accelerated the changes of holding positions and significantly worsened the quality of the rebar market; specifically, the volatility, price impact, and bid-ask spread of rebar went up on the day interventions were published. Moreover, intervention may have attracted short-term speculators that were betting on government actions, rather than investing based on fundamentals. As a result, the connection between the rebar futures market and its spot market practically has broken down since the days interventions were rolled out, the same holds true for the HRC futures and its spot market. Lastly, there is disturbing evidence of informed traders taking unfair advantage in both the rebar and HRC markets, as seen by higher trading volumes and lower open interests for both markets on the day before rebar interventions are announced. Specifically, high trading volume and low open interest at time $t - 1$ are predictive of government intervention at time t . This result is consistent with some traders being “tipped off” so they can act ahead of the government to secure unfair profits at the expense of uninformed traders.

References

- Battalio, R., Mehran, H., Schultz, P., 2012. Market declines: What is accomplished by banning short-selling? *Current Issues in Economics and Finance* 18.
- Boehmer, E., Jones, C.M., Zhang, X., 2013. Shackling short sellers: The 2008 shorting ban. *Review of Financial Studies* 26, 1363–1400.
- Boulton, T.J., Braga-Alves, M.V., 2010. The skinny on the 2008 naked short-sale restrictions. *Journal of Financial Markets* 13, 397–421.
- Brogaard, J., Hendershott, T., Riordan, R., 2017. High frequency trading and the 2008 short-sale ban. *Journal of Financial Economics* 124, 22–42.
- Brunnermeier, M.K., Sockin, M., Xiong, W., 2017. China’s model of managing the financial system.

- Chen, T., Gao, Z., He, J., Jiang, W., Xiong, W., 2019. Daily price limits and destructive market behavior. *Journal of Econometrics* 208, 249–264.
- Draper, P., Fung, J.K.W., 2003. Discretionary government intervention and the mispricing of index futures. *Journal of Futures Markets* 23, 1159–1189.
- Félix, L., Kräussl, R., Stork, P., 2016. The 2011 european short sale ban: A cure or a curse? *Journal of Financial Stability* 25, 115–131.
- Harvey, C.R., Huang, R.D., 2002. The impact of the federal reserve bank’s open market operations. *Journal of Financial Markets* 5, 223–257.
- Lucia, J.J., Pardo, A., 2010. On measuring speculative and hedging activities in futures markets from volume and open interest data. *Applied Economics* 42, 1549–1557. doi:10.1080/00036840701721489.
- Madhavan, A., Richardson, M., Roomans, M., 1997. Why do security prices changes? a transaction level analysis of nyse stocks. *Review of Financial Studies* 10, 1035–1064.
- Naranjo, A., Nimalendran, M., 2000. Government intervention and adverse selection costs in foreign exchange market. *Review of Financial Studies* 13, 453–477.
- Pasquariello, P., 2007. Informative trading or just costly noise? an analysis of central bank interventions. *Journal of Financial Markets* 10, 107–143.
- Pasquariello, P., 2018. Government intervention and arbitrage. *Review of Financial Studies* 31, 3344–3408.
- Peiers, B., 1997. Informed traders, intervention and price leadership: A deeper view of the microstructure of the foreign exchange market. *Journal of Finance* 52, 1589–1614.